Exploratory Data Analysis of Customer Personality and Marketing Behaviour

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Abstract

This project analyses the Customer Personality Analysis dataset to understand how customer characteristics relate to spending behaviour, purchasing activity, and marketing response. The dataset originally contained 2240 customer records and 29 variables. After cleaning missing income values, removing irrelevant columns, converting date fields, and removing unrealistic outliers, the final dataset contained 2236 customers. The analysis uses visualisation, feature engineering, correlation analysis, Principal Component Analysis (PCA), and KMeans clustering. The results show that income, product spending, and purchase activity are closely related, while customers with more children tend to spend less. PCA showed that the strongest pattern in the data was linked to customer value and spending behaviour. KMeans clustering identified three meaningful customer groups of low-spending browsers, high-value premium customers, and regular mid-value customers.

1. Introduction

The dataset used in this project is the Customer Personality Analysis dataset from Kaggle [1]. Each row represents one customer, and the columns describe customer demographics, purchase channels, marketing campaign responses, product spending, and complaints. The original dataset contains 2240 customer records and 29 columns. It is also mostly suitable for numerical analysis because many columns are numerical, such as income, product spending, purchases, web visits, campaign responses, and household counts making it suitable for exploratory data analysis.

The dataset includes demographic variables such as YearBirth, Education, MaritalStatus, and Income. It also includes behavioural variables such as spending on different products, number of purchases through different channels, website visits, campaign acceptance, and complaints [2]. This makes the dataset useful for understanding how customer characteristics relate to spending and marketing behaviour [2].

These variables are suitable for a marketing-focused EDA because demographic information can help explain difference in purchasing power, daily needs, and product preferences. For example, income may affect affordability, while education, marital status, and other household structure may be linked to different customer's lifestyle. Behavioural variables such as product spending, purchase channels, website visits, and campaign acceptance provide direct evidence of how customers interact with the business. Combining these two types of information makes the data set appropriate for exploring both who customers are and how they behave [4].

Table 1: Summary of main data processing steps.

Step	Action
Missing values	Median imputation for Income
Identifier	Removed ID
Constant columns	Removed ZCostContact, ZRevenue
Date conversion	Converted DtCustomer to datetime
Outliers	Removed unrealistic age and income values
Final dataset	2236 customer records

The aim of this project is to explore customer behaviour and identify meaningful customer segments. The analysis focuses on data processing, visualisation, feature engineering, PCA, and K-Means clustering. These techniques are used to summarise the dataset, find important behavioural patterns, and group customers into interpretable segments [3, 13].

2. Data Processing

The raw dataset was first inspected for missing values, duplicate records, irrelevant columns, incorrect data type, and unrealistic values. This was important because later techniques such as PCA and K-Means clustering are sensitive to missing values and extreme numerical values [4].

Table 1 summarises the main data processing steps applied before the main analysis. The initial observation portrayed that the Income column

contained 24 missing values, which was only about 1.1% of the full dataset. Since this was a small proportion, the rows could have been removed. As, income is an important variable for understanding customer spending behaviour, so keeping these records was useful. The missing values were replaced using the median income rather than the mean because income can be affected by very high values, and the median gives a more stable estimate of a typical customer income.

The ID column was removed because it is only a customer identifier and does not describe customer behaviour. The columns ZCostContact and Z_Revenue were also removed because they contained constant values and therefore added no useful information.

Duplicate records were checked before and after removing the unique customer identifier. No exact duplicate records were found in the original raw dataset when the ID column was still present. After removing ID, a second check identified duplicate-like behavioural rows. These were not automatically removed because different customers can share the same recorded demographic and behavioural values once the unique identifier is removed. Therefore, these rows were treated as possible different customers with similar characteristics rather than confirmed duplicate records. This avoids deleting genuine customers and helps preserve the full customer population for PCA and clustering [11].

The Dt _ Customer column was converted into datetime format so that customer tenure could be calculated later. Three customers had unrealistic ages above 100, and one customer had an extreme income value of 666666. These four rows represented only about 0.18% of the full dataset, so removing them did not significantly reduce the data size. They were removed because extreme age and income values could distort visualisations, PCA, and K-Means clustering. After cleaning, the dataset still contained 2236 customer records with no missing values, which was enough for meaningful analysis [3].

3. Data Visualization

Visualisation was used to explore the cleaned dataset before applying PCA and clustering. This section focuses mainly on original cleaned variables such as income, product spending, purchase channels, education level, and campaign response [5]. The purpose is to understand the general shape of the data before creating engineered features and customer segments. All visualisations were created using Matplotlib, and keeping the plotting process transparent and reproducible.

The first variable inspected was customer income because income is likely to influence purchasing power [7]. Figure 1 shows both the distribution and spread of income after cleaning. Most customers are concentrated in the middle-income range, while fewer customers appear at very low or very high income levels. The boxplot confirms that the unrealistic income outlier was removed, although some high-income customers remain. These values were kept because they

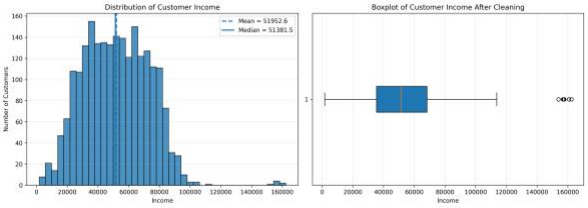


Figure 1: Distribution and boxplot of customer income after cleaning.

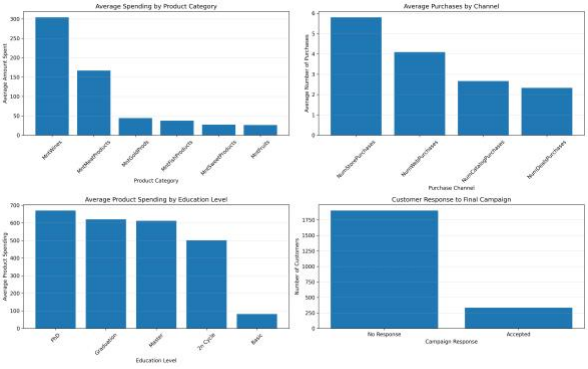


Figure 2: Summary visualisation of raw customer behaviour, including product spending, purchase channels, educationlevel spending, and campaign response.

are plausible and may represent genuine high-value customers.

Income gives an important first view of the customer base, but it does not fully predicts customer behaviour. Customers with similar income may still buy different products, use different purchase channels, or respond differently for marketing campaigns [7]. Therefore, Figure 2 compares product spending, purchase channels, education-level spending, and campaign response to give a broader view of customer behaviour.

Figure 2 shows several useful patterns in the cleaned raw data. Wine and meat products have the highest average spending, suggesting that these product categories are the strongest contributors to customer value. Store and web purchases are also important purchase channels, showing that customers use both physical and online routes. The education-level plot suggests that spending differs across education groups, although this should be interpreted carefully because some categories may contain fewer customers. The

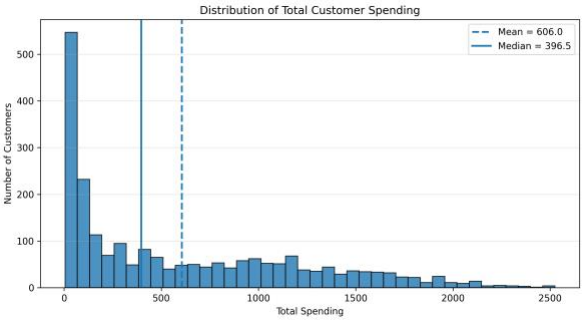
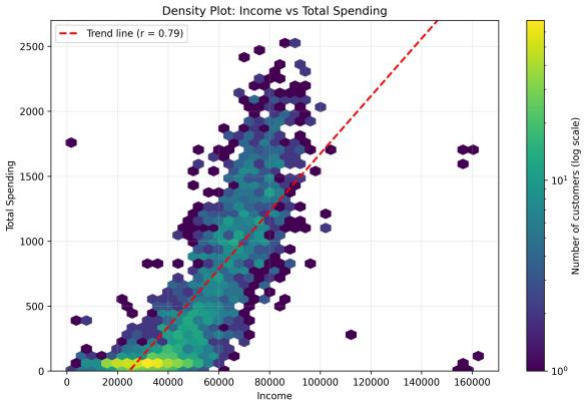


Figure 3: Distribution of engineered total customer spending.



campaign response chart shows that most customers did not accept the final campaign, meaning positive campaign response is relatively uncommon.

Overall, the visualisation stage shows that the dataset contains meaningful variation in income, product preferences, purchase behaviour, and marketing response. These patterns support the feature engineering stage, where related raw variables are combined into clearer customer-level features such as total spending, total purchases, and accepted campaigns.

4. Feature Engineering

Feature engineering was used to create clearer customer-level variables from the raw dataset. The original dataset contains separate columns for product spending, purchase channels, household information, and campaign responses [9]. These were combined into summary features that are easier to interpret and more useful for PCA and clustering [6, 7].

The engineered features were Age, CustomerTenureMonths, TotalChildren, TotalSpending, TotalPurchases, and AcceptedCampaigns. Age was calculated using the latest customer enrolment year in the dataset rather than the current year, so it represents age during the dataset period. CustomerTenureMonths measured how long each customer had been enrolled. TotalSpending combined all product spending columns, while TotalPurchases combined deal, web, catalogue, and store purchases.

These engineered features reduce complexity [8]. Instead of analysing many separate spending and purchase columns individually, it provides direct summaries of customer value, household structure, loyalty, and marketing engagement [9].

Figure 3 shows that total spending is strongly rightskewed. Most customers spend relatively low to moderate amounts, while a smaller group spends much more. This is important because the smaller highspending group may represent a valuable customer segment. These customers were retained because their high spending is meaningful for customer analysis rather than being treated as a data error.

The difference between typical and high-spending customers also shows why segmentation is useful. A single average spending value would hide the fact that the customer base contains very different spending behaviour. This supports the later clustering stage, where customers are grouped according to broader behavioural patterns.

Figure 4 shows a positive relationship between income and total spending. Customers with higher income generally tend to spend more, but there is still variation between customers with similar income levels. This suggests that income is important, but it is not the

Figure 4: Density plot showing the relationship between income and total spending.

only one explanation for spending behaviour. Product preferences, purchase channels, household structures, and campaign response may also affect customer value. Overall, feature engineering helped transform the dataset from separate raw measurements into more meaningful behavioural indicators. These indicators were then used in PCA and clustering to identify broader patterns in the customer base.

5. PCA

Principal Component Analysis (PCA) was used to reduce the dimensionality of the numerical customer features and identify the main directions of variation in the dataset [10, 11]. Since PCA and K-Means clustering are sensitive to feature scale, all selected numerical features were standardised using StandardScaler before PCA and clustering.

Before applying PCA, a correlation heatmap was used to examine relationships between key customer variables. This provides an intermediate step between visualisation and PCA by showing which variables move together [14].

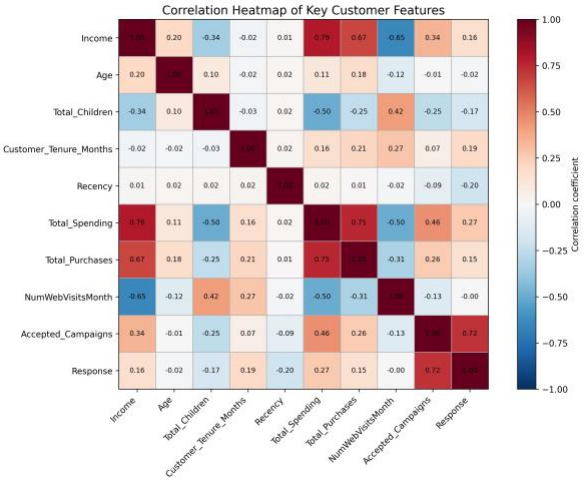


Figure 5: Correlation heatmap of selected customer features.

Table 2: Raw vs engineered PCA comparison.

Feature set	Features	PC1	PC1+PC2	80% PCs
Raw numerical	23	29.11	37.98	12
Engineered summary	9	37.17	51.50	5

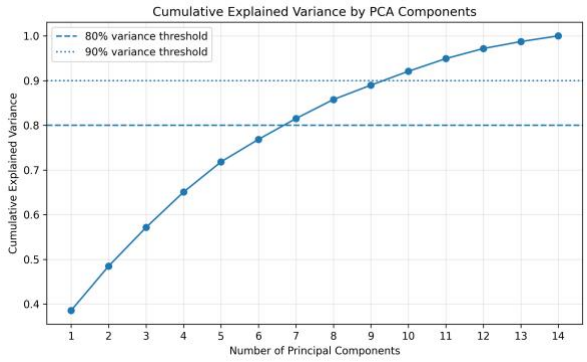


Figure 6: Cumulative explained variance by PCA components.

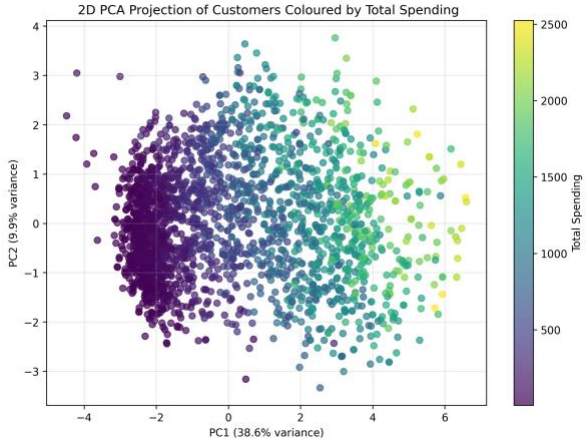


Figure 7: 2D PCA projection of customers coloured by total spending.

Figure 5 shows that income, total spending, and total purchases are positively related. This predicts higherincome customers generally tend to spend and purchase more. The heatmap also shows that customers with more children tend to spend less, and that frequent website visits do not necessarily indicate higher spending. These relationships support the later clustering results, where low-spending customers had more website visits but lower purchase conversion.

Table 2 compares PCA results using raw numerical features and engineered summary features. This comparison was used to test whether feature engineering made the data easier to summarise by reducing repeated information and creating clearer customer level variables.

Table 2 shows that the engineered feature set gives a more compact representation of customer behaviour. With raw numerical features, PC1 explains 29.11% of the variance and the first two components explain 37.98%. With engineered features, PC1 explains 37.17% and the first two components explain 51.50%. The engineered features also require fewer components to preserve 80% of the variance. This suggests that feature engineering helped summarise customer behaviour more effectively.

This comparison also improves interpretability. Raw features contain many separate but related columns, such as individual product spending categories and separate purchase channels. The engineered features combine related information into customer-level measures such as TotalSpending, TotalPurchases, and AcceptedCampaigns. As a result, the main principal components are easier to explain in practical terms, such as customer value, purchase activity, and marketing engagement. This is useful because PCA is not only used to reduce dimensions, but also to understand the main patterns in the dataset.

For the main PCA, TotalSpending was not included because it is created from the individual product spending columns. Including both TotalSpending and the product spending columns would duplicate information. Instead, the main PCA used product category spending columns directly, together with engineered behavioural features.

Figure 6 shows that the first two components are useful for visualisation, but they do not represent the full dataset. Around seven principal components were needed to preserve approximately 80% of the variance, while around ten components were needed to preserve approximately 90%. This means that customer be-

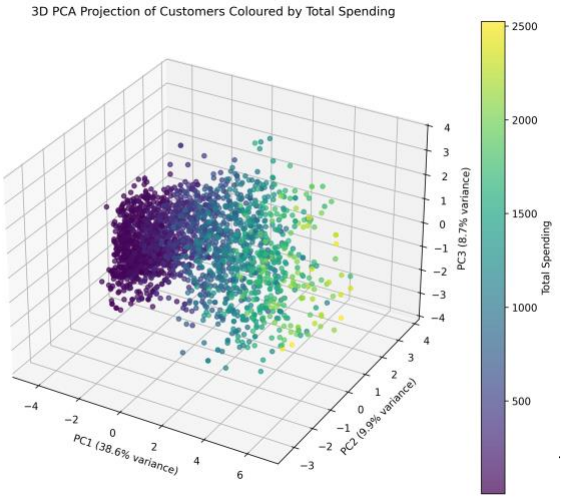


Figure 8: 3D PCA projection of customers coloured by total

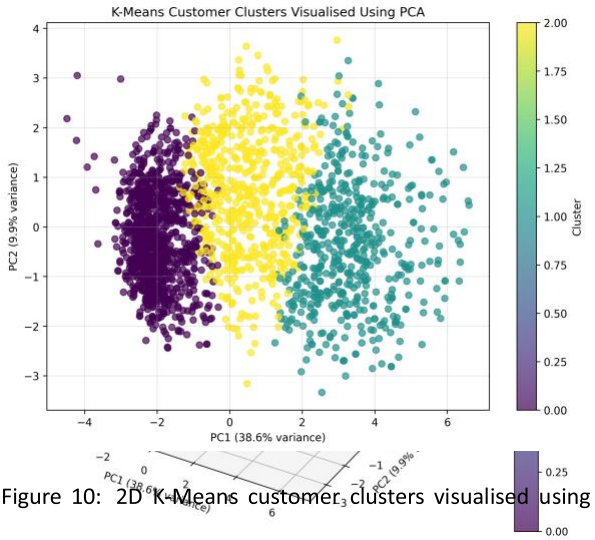


Figure 10: 2D K-Means customer clusters visualised using PCA.

haviour is not explained by only one or two factors; it is influenced by several dimensions.

Figure 7 shows the 2D PCA projection coloured by total spending. A clear spending gradient appears along PC1. Customers with lower spending are mostly on one side, while higher-spending customers are generally further along PC1. This suggests that PC1 represents customer value and spending behaviour.

Although the 2D PCA plot is easy to read, it only shows the first two principal components. Since the cumulative variance plot showed that more components are needed to preserve most of the dataset information, a 3D projection was also produced. This adds PC3 and gives a broader view of the customer structure while still keeping the analysis simpler than the original highdimensional feature space.

Figure 8 adds PC3 to the projection. The main spending pattern still appears most strongly along PC1, but the third component gives additional information about demographic structure. The PCA loadings showed that PC1 was mainly related to income, product spending, and purchases. PC2 was more related to tenure , website visits, purchases, and campaign responses. PC3 was mainly related to age and household structure.

Overall, PCA showed that the strongest pattern in the

6. Clustering

K-Means clustering was used to group customers into behavioural segments based on the same standardised numerical features used for PCA [12, 13, 14]. The purpose of clustering was not to predict a label, but to find natural customer groups based on similar spending, purchasing, and demographic behaviour.

The number of clusters was selected using both the elbow method and silhouette score. The elbow method checks where the reduction in inertia begins to slow down, while the silhouette score measures how clearly separated the clusters are [5].

Figure 9 shows that inertia decreases as the number of clusters increases. The elbow method suggests that improvements begin to slow after around three clusters. The silhouette score is highest for $k = 2$, meaning two clusters give the clearest mathematical separation. However, $k = 3$ was selected because it provides a more useful customer interpretation: low-value, midvalue, and high-value groups.

The silhouette score for the three-cluster solution was modest, indicating that the clusters overlap rather than forming perfectly separated groups. This is not unex-

Figure 11: 3D K-Means customer clusters visualised using PCA.

pected in customer behaviour data, because customers often exist on a spectrum rather than in rigid categories. For example, some customers may sit between mid-value and high-value behaviour, while others may browse frequently without always converting into purchases. Therefore, the clusters should be interpreted as useful exploratory business segments, not as exact natural classes.

Figure 10 shows the three clusters in the first two PCA dimensions. The separation occurs mainly along PC1, which was previously interpreted as the customer value and spending dimension. This suggests that the K-Means model grouped customers mostly according to income, spending, and purchase activity.

The 2D cluster plot is useful because it gives a clear view of the main separation between groups. However, because customer behaviour is influenced by more than two dimensions, the 3D projection was also used

to include PC3 and check whether the segment structure remained visible.

Table 3: Simplified profile of the three K-Means clusters.

Cluster	Income	Spending	Purchases	Campaigns
Low-spending browsers	35052.58	104.21	8.21	0.18
High-value premium	76289.83	1397.46	20.97	0.99
Regular mid-value	59471.96	767.73	20.96	0.43

Figure 11 shows the same clusters in 3D PCA space. Adding PC3 gives an additional view of demographic variation, but the clusters are still not perfectly separated. This agrees with the silhouette score and suggests that the clusters should be interpreted as broad behavioural segments rather than strict customer categories.

Table 3 gives a simplified profile of the three customer segments. Low-spending browsers have the lowest income, lowest total spending, lowest purchase count, and lowest campaign acceptance. This suggests that they browse frequently but do not convert strongly into purchases.

High-value premium customers have the highest income, highest spending, and strongest campaign response. This group is the most valuable segment because they spend substantially more and are more responsive to campaigns.

Regular mid-value customers sit between these two groups. They have medium income and medium spending, but their purchase count is similar to the premium group. This suggests that they buy regularly, but their overall spending is lower than the high-value segment. Overall, the clusters provide useful broad customer segments, even though they should not be treated as perfectly separated groups.

7. Conclusion or Reflection

This exploratory data analysis found that customer spending is strongly right-skewed, with many low- to moderate-spending customers and a smaller group of high-spending customers. Income, total spending, and total purchases were strongly positively related, while customers with more children tended to spend less. Frequent website visits did not necessarily indicate higher spending, suggesting that browsing activity alone is not a strong indicator of customer value.

Feature engineering improved the analysis by transforming raw variables into clearer customer-level features such as TotalSpending, TotalPurchases, TotalChildren, Customer_TenureMonths, and AcceptedCampaigns. The raw vs engineered PCA comparison showed that engineered features provided a more compact representation of customer behaviour and made the principal components easier to interpret.

PCA showed that the main source of variation in the dataset was related to customer value and spending behaviour. PC1 was mainly associated with income, product spending, and total purchases. PC2 captured customer engagement through tenure, website visits, purchases, and campaign response. PC3 was mainly related to age and demographic structure.

K-Means clustering with three clusters identified three interpretable customer segments: low-spending browsers, high-value premium customers, and regular mid-value customers. The silhouette score showed that the clusters were not perfectly separated, which suggests that customer behaviour exists on a spectrum rather than in rigid categories. However, the cluster profiles were still meaningful and useful for exploratory customer segmentation.

Overall, the analysis showed that customer behaviour is influenced by several factors, including income, product preferences, purchase activity, household structure, and campaign response. The main limitation is that this is exploratory analysis, so the results show patterns and associations rather than proving causation. Another limitation is that K-Means assumes roughly spherical clusters, which may not fully match the real shape of customer behaviour. Future work could compare the results with other clustering methods such as hierarchical clustering or DBSCAN, or use predictive modelling to investigate which variables best predict campaign response or high spending.

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